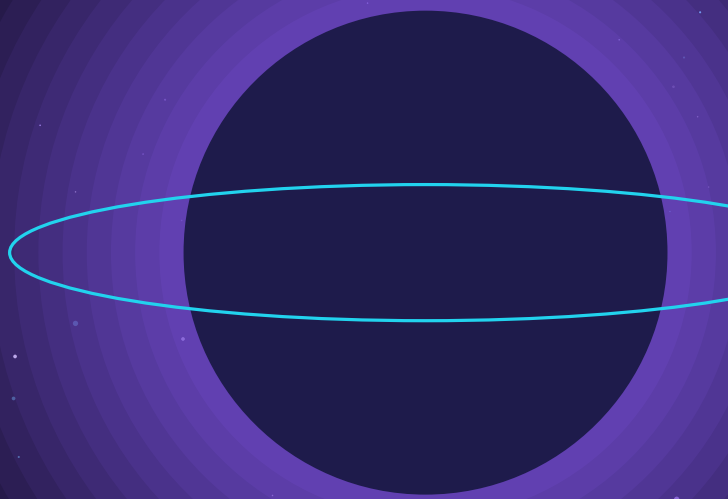
```
> initializing ehsan.exe v2.0 ........ OK
> loading curiosity ................... OK
> welcome, explorer._
```

ehsan.exe

The Level-Up Universe

Portfolio website — complete build plan + Claude Code prompt
with admin panel · Next.js + Neon + Vercel



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Specification v2.0 · October 2026 · 100% free toolchain

0. How this works

This PDF is the complete specification. Ehsan gives it to Claude Code once, in a folder on his computer, and Claude Code builds the **whole website and its admin panel in that session**, phase by phase, saving everything in the folder and committing to Git as it goes. Hosting follows the same proven setup as Ehsan's club website: **GitHub → Vercel** for the site and **Neon** (free Postgres) for the database.

Step	What happens
1	Ehsan creates a folder (e.g. `ehsan-portfolio`), puts this PDF inside, adds a sub-folder `assets_input` with any photo/CV he already has (optional), and connects the folder to Claude Code.
2	Ehsan pastes the **Master Prompt** (Section 13.1). Claude Code reads this PDF and builds Phases 0–9 in order.
3	Claude Code stops only to ask for: the Neon connection string, the admin email/password, and approval before anything is pushed.
4	After each phase Claude Code builds, fixes errors and commits. Progress is tracked in `PROGRESS.md` so the work can always resume.
5	Ehsan tests locally (`npm run dev`), then pushes to GitHub. Vercel deploys automatically; the site reads its content from Neon.
6	From then on: **content and design** are changed in the admin panel; **new features or code changes** are asked from Claude Code in the same folder.

0.1 Code vs content — the key idea

	Lives in	Changed by	Goes live by
Code (animations, components, admin itself)	Your folder → GitHub	Claude Code (you ask in chat)	`git push` → Vercel redeploys (1–2 min)
Content + design (texts, sections, colours, images, effects, achievements…)	Neon database	You, in the admin panel `/admin`	Click **Publish** → live in seconds, no push needed

> **Golden rule:** the website must stay useful even before the magic works. A recruiter who opens it on a slow phone must see your name, role and LinkedIn within 3 seconds. Every effect is layered on top of a fast, readable base — never in front of it.

1. The concept — “ehsan.exe: The Level-Up Universe”

The visitor **boots** Ehsan like a program, witnesses a **Big Bang** that forms his name out of stars, then **travels through space** where every section is a planet. Along the way they discover that Ehsan is a **game character who is levelling up**: skills are stars in a constellation that light up as he learns, projects are quests, and the visitor can unlock **hidden achievements** and open a **secret terminal**.

The theme turns a short student skill list into a strength: unfinished skills glow as “unlocking…”, future goals appear as locked stars. With the admin panel, the universe grows every semester without touching code.

1.1 Where each idea comes from (the fusion)

Original idea	How it lives inside the Level-Up Universe
OS / terminal	Boot sequence “ehsan.exe”, hidden terminal with real commands, HUD in a monospace font.
RPG / game profile	Player card with level and XP bar, skill constellation as a skill tree, projects as quests, achievements.
Space journey	Scrolling = warp travel. Each section is a planet or station. A mini-map shows where you are.
Particle name	Big Bang hero: particles assemble into the name and scatter when touched.
Magic details	Comet cursor, shooting stars, day/night sky by visitor time, Konami code, photo secret, synth sound.

1.2 Personality rules

- Playful but professional: jokes live in micro-copy and Easter eggs, never in the core information.
- Honest: no fake skill percentages. Skills are Mastered / Learning / Locked.
- One “hero effect” per screen. Magic is surprising because it is rationed.

2. Visitor journey — scene by scene

Sections are listed in their default order. **Every text, image, colour and on/off switch below is editable in the admin panel**, and sections can be reordered, hidden, duplicated or added.

Scene 0 — Boot sequence (0 to 2.5 s)

```
> initializing ehsan.exe v2.0 ........ OK
> loading curiosity .................. OK
> compiling dreams ................... OK
> bugs found: 3 · bugs fixed: 3
> welcome, explorer.
```

Lines type out fast, a progress bar fills, then the screen collapses into one bright point. “Press any key / tap to skip” shows from the first frame. Full version on first visit (max 2.5 s), 0.6 s mini version for returning visitors, skipped for `prefers-reduced-motion`. Between 00:00 and 05:00 visitor time one line changes to “it's late and you're here? respect.”

Scene 1 — Big Bang hero

- The point explodes into thousands of particles that assemble into **EHSAN AHMED ABID**; cursor/finger repels them, damped springs pull them back.
- Typewriter taglines: “CSE Student @ East West University”, “Future Software Engineer”, “Bug Hunter”, “Always Levelling Up”.
- **HUD player card** (top-left): mini avatar, “Lv. 2 · CSE Explorer”, gold XP bar, “Achievements 0/10”.
- Top-right: sound toggle (off by default), sky theme toggle, terminal icon. Buttons: “Start the journey”, “LinkedIn”, “Download CV”.
- Name, tagline and buttons are real HTML rendered first; the particle canvas fades in on top.

Scene 2 — Warp travel (background of the whole site)

A fixed 3D starfield behind all sections; scroll velocity stretches stars into light streaks. Each section has a planet or station that drifts in as you arrive. A vertical mini-map (one dot per planet) allows jumping; on mobile it becomes a progress line.

Scene 3 — Planet “Origin” (About)

Photo inside a rotating orbit ring with 3–4 tiny moons (interest icons); 3–5 line story plus one fun line; stat cards (University, Department, Semester, Based in, Currently learning). Clicking the photo 5 times unlocks “Persistent”.

Scene 4 — Skill Constellation (Skills)

- Skills are stars on branches (Programming, Web, CS Fundamentals, Tools, Soft skills — branches editable).
- **Mastered**: bright cyan. **Learning**: violet, pulsing, “unlocking…” ring. **Locked**: dim “?” star with the goal on hover.
- Hover/tap → tooltip: status, where used, since when. An accessible list version sits below the canvas.

Scene 5 — Quest Log (Projects)

Quest cards: title, status (Completed / In progress / Planned), difficulty 1–5, tech tags, “Reward: what I learned”, GitHub/Live buttons, screenshot. 3D tilt with holographic sheen; filter tabs. Lab work and practice apps count.

Scene 6 — Journey Orbit (Timeline)

A curved orbit path whose checkpoints light up while scrolling: school, college, joined EWU CSE, first program, LinkedIn created, website launched, and a locked “???” checkpoint for the future.

Scene 7 — The Lab (Now learning)

A monitoring console with 2–4 current topics, each with a progress meter, a mini oscilloscope sine wave whose frequency rises with progress, and a “last updated” date.

Scene 8 — Signal Station (Contact)

A radio dish emitting rings; form (name, email, message) saved straight into the **admin Inbox** in Neon — no third-party form service. On send: signal-wave animation and “First Contact”. Magnetic LinkedIn / GitHub / Email buttons.

Scene 9 — Edge of the universe (Footer)

“You've reached the edge of the known universe… for now. Ehsan is still levelling up.” Achievement count (x/10), “some secrets are still hidden”, rocket back-to-top, credit line.

3. The magic layer

Feature	Trigger	Behaviour (all parameters editable in admin)
Comet cursor	Mouse (desktop only)	Glowing dot with fading trail; grows on links; magnetic main buttons.
Sky themes	Visitor's local time, toggle or terminal	Dawn · Aurora · Dusk · Deep space, with editable hour ranges and colours. Text contrast never changes.
Shooting stars	Random interval	A star crosses the sky; clicking it unlocks “Wish Granted”.
Achievements	Section 4	Gold toast + chime; HUD counter; progress saved in the visitor's browser.
Hidden terminal	Key ` or ~, or icon	Overlay terminal with history (↑/↓) and Tab completion (Section 5).
God mode	Konami code ↑↑↓↓←→←→BA	10 s of rainbow particles and matrix rain.
Synth sound	Sound toggle (off by default)	Tones generated with the Web Audio API — no audio files.
Smart greeting	Visitor's local hour	“Good morning / evening, explorer”; special boot line at night.

4. Achievements (default set — editable)

#	Name	How to unlock
1	Hello World	First visit
2	Explorer	Scroll to the footer
3	Curious Mind	Open the terminal
4	sudo	Type `sudo hire ehsan`
5	Konami Master	Enter the Konami code
6	Night Owl	Visit between 00:00 and 05:00
7	Persistent	Click the profile photo 5 times
8	Wish Granted	Click a shooting star
9	Stargazer	Inspect 5 skill stars
10	First Contact	Send a message or click the email button

In the admin panel each achievement's name, description, icon and on/off state are editable, and new ones can be added by choosing a trigger type from a fixed list (visit, scroll-to-section, click element N times, terminal command, time window, key sequence, form sent) with its parameter.

5. Terminal commands

Command	Result
help · whoami · about	Command list · short bio
skills · projects · open <name>	Skills by status · quest list · jump to a project
cv · contact · socials	Download CV · email and links

Command	Result
theme night|dawn|aurora · sound on|off	Change sky · toggle sound
achievements · matrix	Unlocked / hidden list · matrix rain
sudo hire ehsan	“Permission granted.” confetti + achievement + opens contact
coffee · joke · clear · exit	Fun replies · clear · close
(unknown)	“command not found — try 'help'. Ehsan is still learning that one too.”

Admins can add custom text commands, edit every reply, and rename or disable built-in commands.

6. Admin panel — “Mission Control”

A private dashboard at `/admin` where Ehsan can change the website from A to Z without code. It is **schema-driven**: the whole site is described by one typed schema, and the admin forms are generated from that same schema. Result: every field the website uses is automatically editable, and future features become editable for free.

6.1 Admin areas

Area	What Ehsan can do
Dashboard	Unpublished-changes counter and **Publish** button, missing-content checklist (every TODO field), unread messages, visits and achievement-unlock counts, last published time.
Sections	List of all sections with drag-and-drop ordering, show/hide, duplicate, delete, rename, nav label, anchor link. **Add new section** from types: About, Skills, Quests, Journey, Lab, Contact, Custom text, Custom cards, Gallery, Video/Link embed, Quote, Call-to-action. Per section: title, subtitle, planet colour/size/ring, background mood.
Content	Profile (name, taglines, bio, location, photo, CV), Skills, Quests, Journey checkpoints, Lab topics, Socials, fun facts and jokes — add, edit, delete, reorder. Rich text with bold, italics, links and lists.
Design	Colour tokens with colour pickers and a **contrast checker** (warns below 4.5:1), theme presets (Deep Space, Nebula, Aurora, Midnight Gold), heading/body/mono fonts from a curated free list, corner radius, glass blur and opacity, button style, cursor style, spacing scale. **Live preview** next to the controls.
Effects	On/off and sliders for: boot (lines, duration, show-once), particles (count per tier, repel force, colours), starfield density, warp intensity, shooting-star interval, sky schedule, sound default and volume, comet trail, Konami, quality-controller thresholds.
Game	HUD labels, level titles, XP rules, achievements (Section 4), terminal commands (Section 5).
Media	Upload images and PDFs; automatic WebP conversion and resizing; required alt text; see where each file is used; replace or delete.
Inbox	Contact-form messages: read/unread, reply by email link, mark spam, delete, export CSV.
SEO & sharing	Page title, description, Open Graph image (upload or auto-generated), favicon, keywords, JSON-LD person data, search-engine indexing on/off.
History	Every publish is saved as a version: preview, compare date, **restore** with one click. Export full backup (JSON + media list) and import it again.
Advanced	Custom CSS box (scoped to the public site, with a warning), maintenance mode (“Ehsan is upgrading the universe”).
Account	Change password, sign out of all devices, view recent login attempts.

6.2 Draft → Preview → Publish

- Edits are saved automatically to a **draft**. Visitors never see half-finished changes.
- **Preview** opens the real website rendering the draft (same components, a “Preview mode” badge).
- **Publish** stores a new version, keeps the last 30, and refreshes the live site cache within seconds.
- **Discard draft** returns to the last published version; **Restore** brings back any older version.

6.3 What the admin cannot do (honest limit)

The admin panel can change all content, layout order, design tokens and every parameter of existing effects. A **brand-new kind** of animation or section layout that does not exist yet is code

— Ehsan asks Claude Code for it in the same folder, and once built it automatically appears in the admin panel thanks to the schema-driven design.

7. Design system (defaults — all editable in admin)

Token	Default	Used for
--bg-deep	#05060F	Page background (deep space)
--panel	rgba(255,255,255,0.05) + blur	Glass cards, HUD, terminal
--cyan	#22D3EE	Primary accent, mastered skills, links
--violet	#8B5CF6	Secondary accent, learning skills, planets
--pink	#F472B6	Highlights, dawn sky
--gold	#FBBF24	XP bar, achievements
--text / --muted	#E5E7EB / #94A3B8	Body / secondary text (contrast ≥ 4.5:1)

- **Fonts:** Space Grotesk (headings), Inter (body), JetBrains Mono (boot, HUD, terminal) — loaded with `next/font` (self-hosted, no layout shift).
- **Motion:** entrances 400–800 ms “expo out”; one big effect at a time; ambient loops slower than 1 Hz; no scroll-jacking.
- **Reduced motion:** no boot, no particles, static stars, simple fades.
- The admin edits these tokens; the public site turns them into CSS variables at render time, so a colour change needs no rebuild.

8. Architecture and tech stack (all free)

Purpose	Tool	Cost
Framework (site + admin + API)	Next.js (latest stable, App Router) + TypeScript	Free
Styling	Tailwind CSS	Free
3D	three.js + @react-three/fiber + @react-three/drei (client-only, lazy-loaded)	Free
Animation / scroll	GSAP + ScrollTrigger, Motion, Lenis	Free
Database	**Neon** serverless Postgres (free plan) + `@neondatabase/serverless`	Free
ORM + migrations	Drizzle ORM + drizzle-kit	Free
Validation	Zod (one schema for site, admin forms and API)	Free
Admin auth	Password hashed with bcrypt, signed httpOnly session cookie (jose)	Free
Admin UI helpers	dnd-kit (drag order), TipTap (rich text), react-colorful (colour picker)	Free
Image processing	sharp (server-side resize + WebP)	Free
Media storage	Stored in Neon, served through a cached route (see 8.2). Optional upgrade: Vercel Blob.	Free
Hosting / deploy	GitHub repo → **Vercel** Hobby (auto-deploy on push)	Free

Purpose	Tool	Cost
Builder	Claude Code with Claude Pro	Already owned

8.1 Database tables

Table	Holds
site_versions	id, status (draft / published / archived), data (JSONB — the entire site: sections, content, design, effects, game, SEO), note, created_at, published_at
media	id, content hash, filename, mime type, bytes, width, height, size, alt text, created_at
messages	id, name, email, message, read, spam, created_at, hashed IP (for rate limiting only)
stats	key → counter (visits, unlocks per achievement); incremented at most once per visitor session
admin_user	email, password hash, session version (for “sign out everywhere”), updated_at
login_attempts	hashed IP, timestamp (brute-force protection)

8.2 Caching — fast for visitors, light on the free tiers

- Public pages are **statically rendered** from the published version and cached with a tag. Publishing calls on-demand revalidation, so visitors almost never trigger a database query. This keeps the site fast and lets Neon's free compute scale to zero most of the time.
- Media URLs contain the file's content hash (`/media/<hash>.webp`) and are served with `Cache-Control: public, max-age=31536000, immutable`, so each image is read from Neon roughly once and then served from Vercel's CDN.
- Uploads are capped (images 4 MB in, resized to max 1920 px, WebP quality ~80, typically under 300 KB; PDFs 5 MB) to protect the free storage.
- The admin and its libraries are route-split and never shipped to public visitors.

9. Engineering targets and optimisation

Metric	Target	How
Name and buttons readable	≤ 1.5 s on 4G phone (boot overlay excluded)	Hero text is static HTML; 3D loads after
Largest Contentful Paint	≤ 2.5 s mobile	Static rendering, next/font, compressed hero photo
Public JS (gzip, first load)	≤ 200 KB	3D in a lazy client chunk; admin code never on public routes
3D chunk (gzip)	≤ 350 KB	Import only used three.js parts
Total first load	≤ 1.5 MB	WebP images, photo ≤ 150 KB
Frame rate	60 fps desktop, ≥ 45 fps mid-range phone	Instanced points, GPU shaders, adaptive tiers
Layout shift (CLS)	≤ 0.05	Reserved space for images and canvas
DB queries per public visit	≈ 0 (cache hit)	Static pages + tag revalidation on publish
Lighthouse mobile	Perf ≥ 85, A11y ≥ 95, Best ≥ 95, SEO 100	Measured in Phase 8

9.1 Adaptive quality — a closed-loop controller

Tier	Hero particles	Background stars	Pixel-ratio cap	Extras
High	8,000	4,000	2.0	All effects, glow
Medium	4,000	2,000	1.5	No glow post-processing
Low	1,500	800	1.0	No warp streaks, no comet trail
Reduced motion	0 (static)	Static image	—	Fades only

- **Initial estimate** from `hardwareConcurrency`, `deviceMemory`, screen size and touch support.
- **Feedback:** average FPS over a 2 s window. Drop one tier if below 40 fps for 2 s; raise only if above 58 fps for 5 s. The asymmetric thresholds (hysteresis) prevent quality flicker. Tier values and thresholds are admin-editable.
- Rendering pauses when the tab is hidden or the canvas is off-screen; without WebGL the site uses the static fallback.

9.2 Accessibility and SEO

- Semantic HTML, one h1, full keyboard navigation, visible focus rings; canvases are `aria-hidden` decoration.
- Metadata, Open Graph image 1200×630 (the LinkedIn preview card), favicon, sitemap.xml, robots.txt, JSON-LD Person.

10. Admin security checklist

- Single admin account. Password stored only as a bcrypt hash; first password comes from `ADMIN_PASSWORD` in `.env.local` during seeding and can be changed in the admin.

- Session: signed httpOnly, Secure, SameSite=Strict cookie, 7-day expiry; “sign out everywhere” bumps the session version.
- **Every** admin server action and API route checks the session on the server — middleware alone is not trusted.
- Login rate limit: 5 failed attempts per 15 min per IP; generic error messages.
- All input validated with Zod; rich text sanitised; uploads checked by real file type, not extension; size caps enforced.
- Contact form: honeypot field, per-IP rate limit, length limits. `/admin` is `noindex`.
- Security headers (CSP, X-Frame-Options, Referrer-Policy). Secrets only in `.env.local` and Vercel environment variables — `.env*` is in `.gitignore` and never committed.

11. Project structure

```
ehsan-portfolio/
├─ ehsan-exe-website-plan.pdf # this specification
├─ CLAUDE.md # condensed spec + rules (written in Phase 0)
├─ PROGRESS.md # phase checklist, updated after every phase
├─ DEPLOY.md / ADMIN_GUIDE.md # deployment steps + how to use the admin
├─ assets_input/ # raw files from Ehsan (optional)
├─ .env.local # secrets (never committed)
├─ drizzle/ # migrations
└─ src/
   ├─ app/
   │ ├─ (site)/page.tsx # public website
   │ ├─ media/[file]/route.ts # cached media delivery
   │ ├─ admin/ # Mission Control (login, dashboard, sections,
   │ │ # content, design, effects, game, media, inbox,
   │ │ # seo, history, advanced, account)
   │ └─ api/ # contact, stats, upload, auth
   ├─ schema/site.ts # Zod schema of the ENTIRE site (single source of truth)
   ├─ db/ (schema.ts, client.ts, seed.ts)
   ├─ lib/ (auth.ts, publish.ts, media.ts, rateLimit.ts, storage.ts, sound.ts, fps.ts)
   ├─ components/
   │ ├─ boot/ hero/ space/ terminal/ achievements/ cursor/ ui/
   │ └─ admin/ (SchemaForm.tsx, ColorField.tsx, MediaPicker.tsx, SortableList.tsx ...)
   └─ sections/ (About, Skills, Quests, Journey, Lab, Contact, Footer, Custom*)
```

The seed (`db/seed.ts`) fills the first published version with all default content, design and effects from this PDF, using `TODO_EHSAN` placeholders where personal details are unknown. The admin dashboard lists every TODO field so nothing is forgotten.

12. Build phases (all in one session, in order)

Phase	Build	Done when
0 · Foundation	Next.js + TS + Tailwind, dependencies, folder structure, CLAUDE.md, PROGRESS.md, .gitignore, git init. Ask for the Neon dev connection string; Drizzle schema + first migration.	`npm run dev` works; tables exist in Neon.
1 · Content model	Zod site schema, seed with all default content, public site rendering every section from the published version (readable, responsive 360–1920 px, no heavy effects yet).	All content readable on phone and desktop from the database.
2 · Admin core	Login/logout, session, rate limit, admin layout, schema-driven forms, Sections manager (drag order, add, hide, duplicate), Content editors, Media library, draft → preview → publish, History/restore, export/import.	Change a text in admin → Publish → visible on the site.
3 · Universe	Starfield, warp streaks, planets, mini-map, sky themes, Lenis + ScrollTrigger reveals — all parameters from settings.	Smooth on desktop; content still readable.
4 · First impression	Boot screen, Big Bang particle name, typewriter taglines, HUD with computed level/XP.	Name readable ≤ 3 s first visit, ≤ 1 s return; skip works.
5 · Signature sections	Skill Constellation (+ list), Quest cards, Journey Orbit, Lab oscilloscopes, Signal Station → Inbox, footer, custom section types.	All sections interactive on mouse and touch.

Phase	Build	Done when
6 · Magic layer	Comet cursor, magnetic buttons, achievements engine (from DB definitions), terminal (DB commands), Konami, secrets, shooting stars, Web Audio.	All achievements unlockable; terminal complete.
7 · Studio	Admin Design (tokens, presets, fonts, contrast checker, live preview), Effects, Game, SEO, Inbox, Advanced (custom CSS, maintenance), Account.	Every design/effect change previewable and publishable.
8 · Optimise + secure	Quality tiers + FPS controller, caching/revalidation check, image pipeline, reduced-motion version, SEO/OG/JSON-LD, accessibility audit, security checklist (Section 10), production build, Lighthouse report.	Targets in Section 9 met; numbers reported.
9 · Deploy prep	DEPLOY.md and ADMIN_GUIDE.md, production seed script, final review. Wait for Ehsan to push.	Ehsan can deploy following DEPLOY.md.

13. Prompts for Claude Code

13.1 Master prompt — paste once, Claude Code builds everything

```
You are a senior full-stack creative engineer. The file
"ehsan-exe-website-plan.pdf" in this folder is the complete specification
for my personal portfolio website with an admin panel. Read the WHOLE PDF
carefully before writing any code.

Goal: build the entire project in this session — Phases 0 to 9 in order
(Section 12) — so it is ready for me to push to GitHub, deploy on Vercel,
and run on Neon Postgres.

Start with:
1. Write CLAUDE.md: a condensed but complete version of the spec
   (concept, scenes, magic layer, achievements, terminal, admin panel,
   design tokens, architecture, DB tables, caching, performance budget,
   quality controller, security checklist, folder structure, phases with
   "done when" criteria, and the rules below).
2. Write PROGRESS.md with a checkbox for each phase.
3. Then build Phase 0 → Phase 9 one after another without waiting for me,
   except for the stop points below.

Only stop and ask me for:
- the Neon connection string for development (Phase 0),
- the admin email and first password (before seeding),
- any dependency that is not listed in the spec,
- final approval before anything is pushed (I push myself).

Rules:
- Free tools only, exactly as listed in the spec. No paid services.
- The Zod site schema is the single source of truth. Every value the
  public site uses must come from the published site version in Neon and
  must be editable in the admin panel. Never hardcode content in components.
- Missing personal info: use TODO_EHSAN placeholders and show them in the
  admin dashboard's missing-content checklist.
- If /assets_input has files, optimise them and add them to the media
  library during seeding.
- Mobile first, prefers-reduced-motion respected, site readable without
  WebGL. Follow the performance budget and security checklist.
- Secrets only in .env.local (gitignored); provide .env.example.
- I am a CSE student: add short comments explaining tricky parts
  (shaders, particle physics, FPS controller, auth, caching).
- After EVERY phase: npm run build, fix all errors and type warnings,
  run a quick self-check against that phase's "done when" criteria,
  commit "phase N: <summary>", tick PROGRESS.md, then continue.
- Keep chat messages short: one line per finished phase.
- At the end: summarise what was built, list remaining TODO_EHSAN items,
  and tell me exactly how to test locally and how to deploy (DEPLOY.md).
```

13.2 If the session stops (usage limit or closed window)

```
Read CLAUDE.md and PROGRESS.md, and check git status and git log.
Continue the build from where it stopped. Do not redo finished phases.
Follow the same rules as before.
```

13.3 Asking for changes later (same folder)

```
Change request: in [section/page], on [phone/desktop], [what happens].
I want [what you expect]. Screenshot attached if useful.
Keep it editable from the admin panel, update the schema if needed,
build, fix errors, commit, and tell me if I need to run a migration.
```

```
New feature: add [feature]. Make all of its texts and settings
editable in the admin panel (schema-driven), follow the performance
budget, and update CLAUDE.md, ADMIN_GUIDE.md and PROGRESS.md.
```

```
It lags on my phone ([model]). Profile it, tune the quality tiers and
FPS controller (Section 9.1), and report before/after numbers.
```

14. Setup, environment and deployment

14.1 Accounts and tools (all free)

- Already done: Git, and the GitHub → Vercel + Neon workflow from the club website.
- Node.js LTS installed; a code editor (VS Code) to look at files.
- Neon: create a new project “ehsan-portfolio”. Create a branch called **dev** for local work; **main** stays for the live site (local experiments never break live content).

14.2 Environment variables

Variable	Value	Where
DATABASE_URL	Neon connection string (pooled). dev branch locally, main branch on Vercel	.env.local and Vercel
AUTH_SECRET	Random 32+ character string (Claude Code generates one)	.env.local and Vercel (different values)
ADMIN_EMAIL	Login email for the admin	.env.local and Vercel
ADMIN_PASSWORD	First password, used only by the seed script — remove after seeding	.env.local (temporarily)
NEXT_PUBLIC_SITE_URL	http://localhost:3000 locally; the live URL on Vercel	.env.local and Vercel

14.3 Deploy (same routine as the club website)

#	Step
1	Test locally: `npm run dev`, open `localhost:3000` and `localhost:3000/admin`.
2	Create an empty GitHub repo and push the folder (Claude Code prepares the commands; you run the push).
3	In Vercel: import the repo (framework auto-detected as Next.js) and add the environment variables from 14.2 using the Neon **main** branch.
4	Run the production migration and seed once (`npm run db:migrate` and `npm run db:seed` with the main-branch URL) — exact commands in DEPLOY.md.
5	Open the live `/admin`, log in, change the password, fill in your real content, press **Publish**.
6	Optional: connect a custom domain in Vercel; update `NEXT_PUBLIC_SITE_URL`.
7	Future code changes: ask Claude Code → test → `git push` → Vercel redeploys automatically.

14.4 Building everything on Claude Pro

- One session can build the whole project, but it is a large job and may reach the Pro usage limit before Phase 9. That is expected and safe: every phase is committed and ticked in PROGRESS.md.
- If it stops, wait for the limit to reset, open Claude Code in the same folder, and paste prompt 13.2 — it resumes exactly where it stopped.
- Fill in texts and images in the admin panel yourself — content editing never uses Claude usage.
- Be specific in change requests (13.3); vague requests like “make it better” cost much more.

15. What Ehsan needs to provide

Nothing blocks the build. Placeholders are used until real content arrives, and **all of this can be typed or uploaded directly in the admin panel after launch**.

Item	Details	When
Neon connection string	dev branch URL for local development	Phase 0
Admin email + password	Strong password (12+ characters)	Before seeding
Profile photo	Square, ≥ 800×800 px, face clear, plain/dark background; transparent PNG looks best	Any time
Short bio	3–5 sentences + one fun line	Any time
Education	EWU start year, current semester, expected graduation; school/college for the timeline	Any time
Skills	Each skill as Mastered / Learning / Locked, and where you used it	Any time
Projects	Name, 1–2 lines, technologies, status, GitHub link, screenshot — small labs count	Any time
Links	LinkedIn URL, GitHub username, public email	Any time
Now learning	2–4 current topics	Any time
Optional	CV PDF, 3–5 fun facts for Easter eggs, clubs, certificates, favourite colour	Any time

16. Launch checklist

- Live site on phone with mobile data: name visible fast, no lag, everything readable.
- Phone “reduce motion” setting on → calm version works.
- Admin: edit a text, change a colour, hide a section, upload an image → Preview → Publish → live within seconds. Restore an old version.
- Send yourself a contact message → it appears in the Inbox.
- Share the link in a private LinkedIn message → preview card shows the OG image and title.
- Add the URL to LinkedIn (Contact info → Website, and the Featured section).
- Ask two friends to hunt the hidden achievements; fix anything confusing.
- Every month: update Skills and Lab in the admin and press Publish. The universe grows with you.

> **Interview tip:** read through the commented code for the particle physics, the starfield shader, the FPS controller, the schema-driven admin and the caching strategy. Being able to explain how your own full-stack site works is worth more than the site itself.